Artificial Intelligence Lab

Dibyasundar Das

10 September 2026

LAB 06

1. Implement a two-player tic-tac-toe game with a line user interface (LUI). Two players share the terminal and enter their moves alternately, one line at a time. The program must display the board after every move, reject invalid or occupied squares without crashing, and announce the winner or a draw as soon as the game ends.
2. Modify your program so that one player is an agent that chooses its moves using the **minimax algorithm**. The agent must compute its move by searching the game tree and not from a lookup table or hand-written rules.

Sample Output :

```
--- TIC-TAC-TOE ---

Do you want to play first? (y/n): y
You are X, the computer is O.

 1 | 2 | 3
---+---+---
 4 | 5 | 6
---+---+---
 7 | 8 | 9

Your move (X): 1

Computer is thinking...
Computer plays 5
 X | 2 | 3
---+---+---
 4 | O | 6
---+---+---
 7 | 8 | 9

Your move (X): 9

Computer is thinking...
Computer plays 2

 X | O | 3
---+---+---
```

```
 4 | O | 6
---+---+---
 7 | 8 | X

Your move (X): 4

Computer is thinking...
Computer plays 8

 X | O | 3
---+---+---
 X | O | 6
---+---+---
 7 | O | X

Computer (O) wins!
```